

Designator	Quantity
D3, D4, D5, D6, D7, D8, D9	7
D1, D2, D10	3
SW1	1
U7	1
R50, R51, R53, R55	4
R1, R2	2
R3, R4, R9, R10, R11, R26,	20
R5, R6	2
R7	1
R8	1
R12, R13	2
R14, R15	2
R16, R17	2
R18, R19	2
R20	1
R21, R24	2
R22	1
R23	1
R25	1
R27	1
R28	1
R29	1
R30, R33, R34, R35, R36, R	9
R37	1
R38, R39, R40, R41, R42, R	9
R48	1
R56	1
R60	1
R66, R69	2
R67, R68, R70, R71, R72	5
NT1	1
Q1	1
L1, L2	2
L3	1
JTAG1, P1	2
P4	1
P5	1
P7	1
P8	1
P10	1
L4	1
X2	1
C45, C101, C104	3

Designator	Quantity
C55, C96, C107, C112	4
C1, C3, C50	3
C2	1
C4	1
C5, C7, C9, C11, C15, C16	22
C6, C8, C10, C12, C13, C14	47
C17, C18, C22, C23	4
C31	1
C33, C34, C40, C53, C65, C66	13
C35, C36	2
C37, C54	2
C38	1
C46	1
C49	1
C56, C57	2
C58, C60, C70, C71	4
C61	1
C102	1
H3, H4, H5, H6	4
P2	1
P3	1
U1, U2	2
U3, U4	2
U5	1
U6	1
U8	1
U9	1
U10	1
U11	1
U12	1
U13	1
U14	1
U15	1
X1	1
X3	1
P6, P9	2
H1, H2	2

Description
LED BLUE CLEAR CHIP SMD
TVS DIODE 16VWM 26VC DO214AA
IC CAN CONTROLLER W/SPI 18SOIC
RES 0.01 OHM 1% 3/4W 1206
RES 105K OHM 1% 1/10W 0603
RES 10K OHM 1% 1/10W 0603
RES 3.32K OHM 1% 1/10W 0603
RES 665 OHM 1% 1/10W 0603
RES 1.33K OHM 1% 1/10W 0603
RES 100K OHM 1% 1/10W 0603
RES 20K OHM 1% 1/10W 0603
RES 24.9K OHM 1% 1/10W 0603
2T WPC2512 3W 50PPM 20 MOHM 1.0%
RES 0.01 OHM 1% 1/3W 0603
RES 102K OHM 1% 1/10W 0603
RES 332K OHM 1% 1/10W 0603
RES 43K OHM 1% 1/10W 0603
RES 49.9K OHM 1% 1/10W 0603
RES SMD 1.3K OHM 1% 1/10W 0603
RES SMD 4.7K OHM 1% 1/10W 0603
RES SMD 120 OHM 1% 1/10W 0603
RES 22 OHM 1% 1/16W 0402
RES 120 OHM 1% 1/8W 0805
RES 1K OHM 1% 1/10W 0603
RES 22 OHM 1% 1/10W 0603
RES 5.1K OHM 1% 1/10W 0603
RES 1.5K OHM 1% 1/10W 0603
MOSFET N-CH 60V 320MA SOT23-3
FIXED IND 8.2UH 3A 106 MOHM SMD
FIXED IND 4.7UH 5.9A 36.3 MOHM
FERRITE BEAD 600 OHM 0603 1LN
2.0MM X 1.2MM TUNING FORK 32.768
CAP TANT POLY 100UF 25V 2917

Description
CAP TANT POLY 22UF 25V 2312
CAP CER 4700PF 50V X7R 0603
CAP CER 3300PF 50V X7R 0603
CAP CER 2200PF 50V X7R 0603
CAP CER 22UF 25V X5R 0805
CAP CER 0.1UF 50V X7R 0603
CAP CER 47UF 25V X5R 1206
CAP CER 1UF 25V X7R 0603
CAP CER 0.1UF 50V X7R 0402
CAP CER 4.7UF 16V X5R 0603
CAP CER 10UF 35V X7R 1206
CAP CER 100PF 50V X7R 0603
CAP CER 22PF 50V C0G/NP0 0603
CAP CER 12PF 50V C0G/NP0 0603
CAP CER 10UF 25V X5R 0603
CAP CER 10UF 35V X5R 0805
IC ELECTRONIC FUSE 10VQFN
IC REG BUCK ADJ 3A 8SOPWR
IC REG BUCK BOOST PROG 4A 21VQFN
IC MCU 8BIT 128KB FLASH 64TQFP
IC TRANSCEIVER HALF 1/1 8SOIC
IC TRANSCEIVER HALF 1/1 8SOIC
IC CURRENT MONITOR 1% 16UQFN
SENSOR ANALOG -40C-125C SOT23-3
IC TRANSCEIVER FULL 2/2 16TSSOP
IC OSC PROG TIMER SOT-23-6
IC EEPROM 4KBIT I2C 400KHZ 8SOIC
IC REDRIVER I2C HOTSWAP 8TSSOP
CRYSTAL 16.0000MHZ 8PF SMD
CRYSTAL 8.0000MHZ 8PF SMD
CONN SOCKET 52POS 0.1 GOLD PCB

Footprint	Name	Manufacturer 1
Led 0603	Led	Diodes Inc.
DO-214AA_SMB	SMBJ16A	Taiwan Semiconduct
SW1-SMD	SW2	C&K Components
SOIC-20	MCP2515-SO	Microchip
RES_C_1206_SHUNT	Shunt 4 Pin	Susumu
RES_C_1608X60	105k	Yageo Group
RES_C_1608X60	10k	Yageo Group
RES_C_1608X60	3.32k	Yageo Group
RES_C_1608X60	665R	Yageo Group
RES_C_1608X60	1.33k	Yageo Group
RES_C_1608X60	100k	Yageo Group
RES_C_1608X60	20k	Yageo Group
RES_C_1608X60	24.9k	Yageo Group
RES_C_3216X70	DNP/4.7k	
RES_C_6331X65_N	20mR	Littelfuse
RES_C_1608X60	10mR	SEI
RES_C_1608X60	102k	Yageo Group
RES_C_1608X60	332k	Yageo Group
RES_C_1608X60	43k	Yageo Group
RES_C_1608X60	49.9k	Yageo Group
RES_C_1608X60	DNP/150k	
RES_C_1608X60	11.3k	Bourns
RES_C_1608X60	4.7k	Bourns
RES_C_1608X60	120R	Bourns
RES_C_1005X40	22R	Yageo Group
RES_C_2012X65	120R	Yageo Group
RES_C_1608X60	1k	Yageo Group
RES_C_1608X60	22R	Yageo Group
RES_C_1608X60	5.1k	Yageo Group
RES_C_1608X60	1.5k	Yageo Group
NETTIE_SMD	NT	
SOT-23	2N7002	onsemi
INDUCTOR 6363	8.2uH	Vishay
Inductor SMD 6x5.7x5	4.7uH	Eaton
HDR/Header/4/2mm	Header 4	
HDR/Header/2/BOX/No STEP	+12V IN	
HDR/Header/2/BOX/No STEP	+12V TEST	
HDR/Header/2/BOX/No STEP	+VOUT TEST	
HDR/Header/2/BOX/No STEP	+5V_TRANS TEST	
HDR/Header/2/BOX/No STEP	+5V TEST	
L 0603 (1608)	FB 600R	Murata
CRYSTAL_NX2012SA	Crystal 2 Pins	KYOCERA AVX
C SMD TAN 2917 (E)	100uF	Vishay

Footprint	Name	Manufacturer 1
C SMD TAN 2312 (C)	22uF	KYOCERA AVX
CAP_C_1608X80	4.7nF	Samsung Electro-Me
CAP_C_1608X80	3.3nF	Samsung Electro-Me
CAP_C_1608X80	2.2nF	Samsung Electro-Me
CAP_C_2012X110	22uF	KYOCERA AVX
CAP_C_1608X80	100nF	Yageo Group
CAP_C_3216X140	47uF	TDK
CAP_C_1608X80	DNP/100nF	
CAP_C_1608X80	1uF	TDK
CAP_C_2012X110	DNP/100nF	
CAP_C_1005X60	100nF	Samsung Electro-Me
CAP_C_1608X80	4.7uF	TDK
CAP_C_3216X140	10uF	Samsung Electro-Me
CAP_C_1608X80	100pF	Yageo Group
CAP_C_1608X80	22pF	Yageo Group
CAP_C_1608X80	12pF	Yageo Group
CAP_C_1608X80	10uF	KYOCERA AVX
CAP_C_2012X110	10uF	TDK
HEX-M3	Shield	
0532610471	UART_DEBUG	Molex
0532610471	UART_CONSOLE	Molex
TPS259474	TPS259470ARPWR	Texas Instruments
HSOIC_8	LMR33630ADDA	Texas Instruments
VQFN-HR-21	TPS55287RYQR	Texas Instruments
TQFP-64	ATMEGA128A-AU	Atmel
SOIC-8	SN65HVD251DR	Texas Instruments
SOIC127	MAX485ESA+	Analog Devices
QFN65	PAC1934T-I/JQ	Microchip
SOT-23	TC1047AVNBTR	Microchip
TSSOP-16	SN65C3232PWR	Texas Instruments
SOT-23-THN-6	TPL5010QDDCTQ1	Texas Instruments
SOIC_127	24LC04BT-E/SN	Microchip
TSOP8	PCA9508DP,118	NXP Semiconductors
XTAL_2520	16Mhz	NDK
XTAL_3225	8Mhz	ECS International
532610671	532610671	Molex
ESQ-126-14-G-D	ESQ-126-14-G-D	Samtec

Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1
LTST-C191TBKT	Volume Production	Mouser	859-LTST-C191TBKT
SMBJ16A	Volume Production	Mouser	821-SMBJ16A
PTS636SK25FSMTRLFS	Volume Production		
MCP2515T-E/SO	Volume Production	Mouser	579-MCP2515T-E/SO
KRL1632E-M-R010-F-T5	Volume Production	DigiKey	408-1570-1-ND
RC0603FR-07105KL	Volume Production	Newark	RC0603FR-07105KL
RC0603FR-0710KL	Volume Production	Mouser	603-RC0603FR-0710KL
RC0603FR-073K32L	Volume Production	Newark	RC0603FR-073K32L
RC0603FR-07665RL	Volume Production	Mouser	603-RC0603FR-07665RL
RC0603FR-071K33L	Volume Production	Mouser	603-RC0603FR-071K33L
RC0603FR-07100KL	Volume Production	Mouser	603-RC0603FR-07100KL
RC0603FR-1320KL	Volume Production	Mouser	603-RC0603FR-1320KL
RC0603FR-0724K9L	Volume Production	Mouser	603-RC0603FR-0724K9L
WPC2512QLR020FYR	Volume Production	Arrow Electroni	WPC2512QLR020FYR
CSM0603FT10L0	Volume Production	Mouser	708-CSM0603FT10L0
RC0603FR-07102KL	Volume Production	Mouser	603-RC0603FR-07102KL
RC0603FR-07332KL	Volume Production	Mouser	603-RC0603FR-07332KL
RC0603FR-1343KL	Volume Production		
RC0603FR-0749K9L	Volume Production	Mouser	603-RC0603FR-0749K9L
CR0603-FX-1301ELF	Volume Production	Mouser	652-CR0603FX-1301ELF
CR0603-FX-4701ELF	Volume Production	Mouser	652-CR0603FX-4701ELF
CR0603-FX-1200ELF	Volume Production	Newark	CR0603-FX-1200ELF
RC0402FR-0722RL	Volume Production	Mouser	603-RC0402FR-0722RL
RC0805FR-07120RL	Volume Production	Mouser	603-RC0805FR-07120RL
RC0603FR-071KL	Volume Production	Mouser	603-RC0603FR-071KL
RC0603FR-0722RL	Volume Production	Mouser	603-RC0603FR-0722RL
RC0603FR-075K1L	Volume Production	Newark	RC0603FR-075K1L
RC0603FR-071K5L	Volume Production	Mouser	603-RC0603FR-071K5L
2N7002KT1G	Volume Production	Newark	2N7002KT1G
IHLP2525BDER8R2M01	Volume Production	Arrow Electroni	IHLP2525BDER8R2M01
EXL1V0503-4R7-R	Volume Production	Arrow Electroni	EXL1V0503-4R7-R
BLM18KG601SN1D	Volume Production	Mouser	81-BLM18KG601SN1D
DT2012SB32768H5HPWAA	Volume Production	Mouser	581-DT2012SB32768H5H
T55D107M025C0060	Volume Production	Mouser	74-T55D107M025C0060

Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1
TCJC226M025R0100	Volume Production	Mouser	581-TCJC226M025R0100
CL10B472KB8NNNC	Volume Production	Mouser	187-CL10B472KB8NNNC
CL10B332KB8NNNC	Volume Production	Mouser	187-CL10B332KB8NNNC
CL10B222KB8NNNC	Volume Production	Mouser	187-CL10B222KB8NNNC
KGM21AR51E226MU	Volume Production	Mouser	581-KGM21AR51E226MU
C0603C104K5RACTU	Volume Production	Mouser	80-C0603C104K5R
C3216X5R1E476M160AC	Volume Production	Mouser	810-C3216X5R1E476M
C1608X7R1E105K080AE	Volume Production	Mouser	810-C1608X7R1E105KAE
CL05B104KB5NNNC	Volume Production	Mouser	187-CL05B104KB5NNNC
C1608X5R1C475K080AC	Volume Production	Mouser	810-C1608X5R1C475KAC
CL31B106KLHNFNE	Volume Production	Mouser	187-CL31B106KLHNFNE
C0603C101K5RACTU	Volume Production	Mouser	80-C0603C101K5R
C0603C220G5GACTU	Volume Production	Mouser	80-C0603C220G5G
C0603C120J5GACTU	Volume Production	Mouser	80-C0603C120J5G
06033D106KAT2A	Volume Production	Mouser	581-06033D106KAT2A
C2012X5R1V106K125AC	Volume Production	Newark	C2012X5R1V106K125AC
53261-0471	Volume Production	Mouser	538-53261-0471
53261-0471	Volume Production	Mouser	538-53261-0471
TPS259470ARPWR	Volume Production	Mouser	595-TPS259470ARPWR
LMR33630ADDA	Volume Production	Arrow Electroni	LMR33630ADDA
TPS55287RYQR	Volume Production	Mouser	595-TPS55287RYQR
ATMEGA128AAU	Volume Production	Newark	ATMEGA128A-AU
SN65HVD251DR	Volume Production	Newark	SN65HVD251DR
MAX485ESA+	Volume Production	Newark	MAX485ESA+
PAC1934TI/IJQ	Volume Production	Arrow Electroni	PAC1934T-I/IJQ
TC1047AVNBTR	Volume Production	Arrow Electroni	TC1047AVNBTR
SN65C3232PWR	Volume Production	Mouser	595-SN65C3232PWR
TPL5010QDDCTQ1	Volume Production	Mouser	595-TPL5010QDDCTQ1
24LC04BT-E/SN	Volume Production	Arrow Electroni	24LC04BT-E/SN
PCA9508DP,118	Volume Production	Newark	PCA9508DP,118
STDCSW5-16M	Unknown	Mouser	344-STDCSW5-16M
ECS-80-8-33Q-ADS	Volume Production	Mouser	520-ECS-80-8-33Q-ADS
53261-0671	Volume Production	Newark	53261-0671
ESQ-126-14-G-D-LL	Volume Production		

Supplier Unit Price 1	Supplier Subtotal 1
0.17	1.7
0.61	1.83
2.97	2.97
0.43	1.72
0.101	0.202
0.014	0.28
0.101	0.202
0.1	0.1
0.12	0.12
0.014	0.14
0.014	0.14
0.014	0.14
0.2033	0.2033
0.66	1.32
0.1	0.1
0.1	0.1
0.1	0.1
0.1	0.1
0.012	0.12
0.1	0.1
0.009	0.09
0.1	0.1
0.1	0.1
0.1	0.1
0.101	0.202
0.014	0.14
0.214	0.214
0.7176	1.44
0.3453	0.3453
0.1	0.1
0.63	0.63
2.41	7.23

Supplier Unit Price 1	Supplier Subtotal 1
1.96	7.84
0.16	0.48
0.2	0.2
0.1	0.1
0.653	14.37
0.048	2.26
0.94	3.76
0.107	1.39
0.14	0.28
0.24	0.24
0.71	0.71
0.12	0.12
0.39	0.78
0.035	0.35
0.5	0.5
0.416	0.416
0.69	0.69
0.69	0.69
1.72	3.44
2.88	5.77
5.32	5.32
7.08	7.08
4.15	4.15
6.41	6.41
1.88	1.88
0.7759	0.7759
3.31	3.31
1.79	1.79
0.4058	0.4058
2.32	2.32
1	1
0.87	0.87
0.766	1.53